



AEGIS Architecture

Anonymization & Exchange Gateway for Imaging Studies

SENDING SOURCES External Browser Upload · DICOM Network (C-STORE SCP, port 11112) · Programmatic Batch Ingest

React Upload Portal (PWA) • Drag-and-drop: files, folders, DICOM directories • Client-side tag de-identification (DICOM PS3.15) • Anonymization preview — before / after tag diff • Multi-study detection with per-study upload cards • Per-project retained-tag anonymization profiles • Auto-retry: 3x exponential backoff per file	DIMSE Receiver (pynetdicom · GCE VM) • Compute Engine VM — aegis-prod-dimse-receiver • Static IP 35.232.172.221 · TCP port 11112 • C-STORE SCP — receives from PACS systems / scanners • Institution attribution (AE title or IP CIDR) • Calls POST /api/ingest on DICOM association close • Durable retry queue + dead-letter (disk-persistent)	Key Privacy Principle PHI is stripped in the browser BEFORE data leaves the hospital. • Only tag-de-identified DICOM is transmitted • HIPAA Safe Harbor — 18 identifier types removed • Deterministic UID hashing + date shifting • Zero software install required at sending site • DIMSE path ingests raw files (internal-only path)
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GCP PROJECT aegis-prod-488120 · us-central1 · Live, February 2026

Cloud Armor (DDoS / WAF) + Global HTTPS Load Balancer + Identity-Aware Proxy (admin routes) + Managed SSL cert v3 (api · admin · aegisimaging.ai · www) + Cloud Run (API · Dashboard · Landing · 8 sidecars)		
API Backend (Go / Cloud Run) • Upload orchestration (sessions, chunked PUT) • Study / project / institution management • Routing rules engine → auto-pipeline dispatch • DICOMweb proxy (QIDO-RS + WADO-RS) • Export shares: token-auth, ZIP download • Email digest scheduler + SMTP relay • DIMSE retry control proxy • MCP tool backend + batch import CLI • distroless image — minimal CVE surface	Admin Dashboard + Weasis DWV (React / Cloud Run) • Behind Identity-Aware Proxy (IAP) • Study browser: filter, search, paginate, bulk ops • Weasis DWV — yoked before/after defacing review • 7-stage pipeline visualization per study • RBAC: admin (write) + viewer (read-only) • Routing rules, institutions, anon profiles • Protocol templates, API keys, webhooks • Audit log + CSV export, share management • Federation peers, project lifecycle	Cloud SQL (PostgreSQL 15) • Studies, projects, institutions, admin users • Routing rules, audit trail, export shares • Private IP · Secret Manager credentials • Point-in-time recovery (7-day retention) Cloud Storage (GCS) • dicom/raw/{uid}/ — tag-de-identified • dicom/clean/{uid}/ — defaced + processed • bids/{uid}/ — NIFTI / BIDS output • Shared volume (Go API + all 8 sidecars)

Processing Sidecars (Python / Cloud Run) — dispatched async by Go API, runs hands-free via routing rules

Defacing (Python) • Head MRI / PET / CT — facial feature removal • mri_reface / DeepDefacer / mri_deface backends • nibabel fallback for dev / testing only • Writes defaced DICOM to clean/ store • SSIM-based QA score per study	PHI Detection (Python) • Burned-in text OCR on DICOM pixel data • Tesseract (local) / Cloud Vision / Textract • Per-file findings with confidence scores • Per-project confidence threshold config • Informational — non-blocking to pipeline	QC Automation (Python) • File integrity + required DICOM tag check • Slice consistency (rows / cols / pixel spacing) • SNR estimation (signal mean / corner noise) • Coverage completeness by body part • Missing slice gap detection	Classification (Python) • Fills modality + body_part from DICOM headers • Heuristic: tags → SOP UID → series description • Cloud Vision / Rekognition image fallback • Re-evaluates routing rules after classify • Confidence threshold gating (≥ 0.5)
BIDS Conversion (Python) • DICOM → NIFTI via dcm2nix • BIDS-compliant directory structure • sub-{hash8}/anat func dwi perf pet/ • JSON sidecar metadata per series • Privacy: UID-hashed subject labels • ZIP download via Go API	Protocol Check (Python) • Verifies TR / TE / flip / thickness vs template • Classic + Enhanced DICOM (multi-frame) • Per-project protocol templates with CRUD • numeric / exact / range / contains_all match • critical / warning / info severity levels	Synth MRI (Python) ★ • Synthetic brain MRI generation • CPU: Shepp-Logan phantom (numpy + pydicom) • GPU: MONAI BraTS LDM (INCLUDE_MONAI=true) • T1w contrast, Rician noise, face anatomy • Seeded + reproducible, 20 – 200 slices • Imports via normal pipeline (auto-dispatch)	
Security Layers • VPC private subnets + Cloud NAT • Cloud Armor DDoS / WAF • TLS 1.2+ on all endpoints • CMEK (Cloud KMS) • IAM least privilege • Secret Manager (DB creds, API keys) • Cloud Audit Logs • Artifact Registry vuln scanning • distroless containers (Go API) • IAP on all admin routes • No PHI in email / audit entries	Export & Sharing Export Portal (React / public SPA) • Token-authenticated share links • Study info, expiry countdown (server-anchored) • ZIP download of approved DICOM files DICOM Forwarding (route_to) • DICOMweb STOW-RS destinations • DIMSE C-STORE to remote AE Title • Auto-forwards on study approval Share Lifecycle • Extend / revoke anytime • Immutable download log per share • Export analytics dashboard	MCP Server + AI Agent Tools Model Context Protocol (Claude) • 17 read tools: studies, audit, stats, health • 13 write tools (confirm:true + reason required) • Zod-validated schemas — read-safe by default • DIMSE retry proxy: process, replay, dead-letter Batch Import + Webhooks + API Keys • aegis-import CLI — bulk historical migration • 5 webhook events, HMAC-SHA256 signed payloads • Machine-to-machine bearer tokens (hashed)	Landing Page (nginx / Cloud Run) • aegisimaging.ai — public marketing site • React + Vite built, served by nginx • Interactive synthetic DICOM demo widget • Schedule Demo / Contact form • Market opportunity: \$45B imaging market • LB default backend — no IAP required • www + apex domains on SSL cert v3

AWS ACCOUNT 301691475234 · us-east-1 · Deploying Q1 2026 — same app layer as GCP, parallel infrastructure

ALB + Cognito (Auth Layer) • HTTPS listener on ACM certificate • Cognito hosted UI — admin-create-only user pool • authenticate-cognito default action • Public bypass rules: /healthz, upload, export	API + Admin (ECS Fargate) • Go API — same image as GCP (1 vCPU / 2 GB) • Admin Dashboard — React / nginx (0.5 vCPU / 1 GB) • Service discovery: api.aegis.local • Force-new-deployment via GitHub Actions	7 Python Sidecars (ECS Fargate) • defacing · phi-detection · qc-service • bids-service · classification-service • protocol-service · synth-service • Cloud Map private DNS: svc.aegis.local:8080
RDS + S3 (Data Layer) • RDS PostgreSQL 15 — private subnet • Secrets Manager — master credentials • S3 DICOM bucket — versioned, KMS-encrypted • Same STORAGE_MODE=s3 as GCP (same Go code)	DIMSE EC2 (t3.small) • Elastic IP — stable for PACS AE title registration • Amazon Linux 2023 — Docker + SSM agent • SSM Parameter Store → image URI on every boot • GitHub Actions: write SSM param + reboot instance	GitHub Actions CI/CD • Triggers on push to develop (same as GCP Cloud Build) • Matrix build: 10 services, --platform linux/amd64 • Push SHA tag + latest tag to ECR • Force-new-deployment for 9 ECS services • Update SSM param + reboot DIMSE EC2

Automated Processing Pipeline routing rules define which steps are required → pipeline dispatches each step in dependency order, hands-free

1. Classify fills modality body_part	2. PHI Scan OCR pixels flag burned-in text	3. Protocol TR / TE / flip vs template	4. Deface remove face head imaging only	5. QC Check SNR · coverage slice consistency	6. BIDS NIFTI + sidecar BIDS structure	7. Export Fwd STOW-RS or DIMSE forward
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Implementation Phases				
Phase 1: Foundation ✓ Terraform (GCP + AWS) · Go API · Upload Portal · Admin Dashboard · PostgreSQL · CI/CD	Phase 2: Processing ✓ Defacing · PHI Detection · QC · BIDS · Classification · Protocol · DIMSE Receiver	Phase 3: Operations ✓ Routing rules · Institutions · Audit · Export Portal · Shares · Email · MCP Server	Phase 4: Production ✓ GCP live (aegis-prod) · DIMSE on GCE VM · Cloud Build CI/CD · IAM hardening · Landing Page · RBAC	Phase 5: Beta → GA 🚧 Private beta · AWS deployment (Q1, deploying now) · Azure + SOC 2 (Q3) · Enterprise GA (Q4)

Key Open-Source Dependencies Go: suyashkumar/dicom · pgx · testcontainers-go · testify Browser: dcmjs · dicomParser · Weasis DWV (MIT) · React 19 · Vite	Multi-Cloud + Test Coverage GCP: Cloud Run · Cloud SQL · GCS · IAP · Cloud Armor · KMS · Pub/Sub AWS: ECS Fargate · RDS · S3 · ALB + Cognito · ACM · Secrets Manager
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Defacing: [mri_reface](#) · [DeepDefacer](#) · [mri_deface](#) · [dcm2niix](#) · [nibabel](#)
PHI/OCR: [pytesseract](#) · [Google Cloud Vision](#) · [AWS Textract](#) · [Pillow](#)
QC/BIDS: [pydicom](#) · [numpy](#) · [dcm2niix](#) · [pynetdicom](#) (C-STORE SCP)
Synth: [numpy](#) · [pydicom](#) · [MONAI Generative](#) · [torch](#) · [huggingface-hub](#)
Infra: [Terraform Google + AWS providers](#) · [Docker distroless/slim](#)

Local: [Docker Compose](#) · [PostgreSQL 15](#) · [Mailpit](#) · [local filesystem](#)
Auth: [GCP IAP](#) · [Azure AD Easy Auth](#) · [AWS ALB+Cognito](#) · [dev auto-auth](#)
Storage: [STORAGE_MODE=gs](#) | [s3](#) | [local](#) — [same Go API](#), no code changes
CI/CD: [GitHub Actions: Go \(120+\)](#) · [Python \(252+\)](#) · [TS](#) · [Docker \(9\)](#) · [Cloud Build auto-deploy](#)
Domains: [aegisimaging.ai](#) · [www](#) · [api](#) · [admin](#) — [SSL cert v3](#)